

SHANTOU, China

WMO: 593160

Lat: 23.3853N

Lon: 116.6792E

Elev: 3

StdP: 101.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	8.8	-1.2	3.4	12.6	1.7	4.3	12.2	8.9	14.6	8.0	14.5	2.2	20	0.341

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	34.9	27.0	33.9	27.0	32.9	26.9	28.6	31.8	28.2	31.1	27.7	30.6	3.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.1	30.4	27.2	23.0	29.6	27.1	22.8	29.5	93.2	31.9	91.0	31.4	89.0	31.0	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.0	6.1	DB	4.6	36.8	1.7	0.9	3.4	37.5	2.4	38.1	1.5	38.6	0.3	39.3	(4)
			WB	2.2	29.6	2.3	0.7	0.6	30.2	-0.8	30.6	-2.1	31.0	-3.8	31.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	15.1	15.7	18.0	22.0	25.5	28.1	29.3	29.1	28.0	25.1	21.2	16.8	(6)
		DBStd	5.76	2.92	3.38	3.23	3.13	2.52	1.89	1.53	1.54	1.69	2.09	2.70	2.93	(7)
		HDD10.0	4	2	2	0	0	0	0	0	0	0	0	0	1	(8)
		HDD18.3	314	106	85	47	7	0	0	0	0	0	0	7	63	(9)
		CDD10.0	4699	159	162	247	361	480	542	597	593	541	468	337	212	(10)
		CDD18.3	1967	6	12	35	117	222	292	339	335	291	209	94	17	(11)
		CDH23.3	19024	12	35	121	587	1737	3192	4178	4020	3179	1583	352	28	(12)
(13)	Wind	CDH26.7	6879	1	3	17	131	483	1149	1824	1691	1161	375	44	1	(13)
		WSAvg	2.6	2.6	2.7	2.7	2.6	2.5	2.6	2.6	2.5	2.6	2.8	2.6	2.6	(14)
		PrecAvg	1557	34	51	83	149	198	300	213	246	148	49	42	32	(15)
		PrecMax	2509	249	320	413	602	592	721	486	650	550	252	233	132	(16)
		PrecMin	927	0	1	6	3	7	36	38	27	16	0	0	0	(17)
		PrecStd	369	39	65	85	116	121	155	121	154	104	65	52	34	(18)
		DB	25.0	26.9	28.8	31.2	34.1	35.1	36.1	36.0	34.9	32.9	29.7	26.0		(19)
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	19.2	21.1	22.8	24.9	27.0	26.9	27.3	27.2	26.3	24.9	23.1	20.3		(20)
		DB	22.2	24.1	26.2	29.8	32.0	33.4	34.8	34.6	33.2	31.0	28.0	23.9		(21)
		MCWB	17.4	19.8	21.6	24.4	26.6	27.2	27.3	27.1	26.3	24.4	22.3	19.1		(22)
		DB	20.4	22.0	24.4	28.1	30.7	32.2	33.6	33.3	32.2	29.9	26.8	22.2		(23)
		MCWB	16.1	18.3	20.4	23.7	26.0	27.0	27.2	27.1	25.9	23.7	21.5	17.8		(24)
		DB	19.1	20.1	22.8	26.8	29.2	31.1	32.5	32.2	31.2	28.8	25.2	21.1		(25)
		MCWB	15.5	17.0	19.2	23.1	25.2	26.9	27.0	26.8	25.6	23.0	20.7	17.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	20.2	22.7	23.9	26.3	28.2	28.9	29.3	29.2	28.6	26.6	24.7	21.9		(27)
		MCDB	23.6	24.3	27.0	29.1	30.8	31.5	32.9	32.6	31.7	29.8	27.4	24.3		(28)
		WB	18.6	20.6	22.6	25.3	27.3	28.2	28.6	28.5	27.7	25.9	23.7	20.4		(29)
		MCDB	21.1	23.3	25.3	28.1	30.2	30.8	32.0	31.8	30.8	28.9	26.2	22.7		(30)
		WB	17.3	19.0	21.4	24.5	26.6	27.8	28.1	28.0	27.1	25.3	22.8	19.3		(31)
		MCDB	19.6	21.2	23.5	27.2	29.4	30.3	31.2	31.1	30.1	28.1	25.3	21.5		(32)
		WB	16.2	17.6	20.0	23.6	26.0	27.4	27.6	27.5	26.7	24.6	21.9	18.1		(33)
(34)	Mean Daily Temperature Range	MCDB	18.5	19.8	22.1	26.1	28.5	29.9	30.7	30.5	29.5	27.4	24.5	20.5		(34)
		MDBR	7.0	6.6	6.7	6.4	5.9	5.6	6.3	6.4	6.7	7.2	7.4	7.5		(35)
		5% DB	8.8	9.5	9.6	8.4	7.8	7.1	7.9	7.9	7.8	8.0	8.2	8.4		(36)
		MCDBR	4.6	5.3	5.3	4.2	3.6	2.8	3.0	3.1	3.1	3.3	3.8	4.4		(37)
		5% WB	7.5	8.6	8.5	7.7	7.0	6.1	6.7	6.8	6.8	6.5	7.0	7.0		(38)
		MCWBR	4.5	5.6	5.8	4.4	3.8	3.1	3.3	3.3	3.2	3.2	3.8	4.2		(39)
		taub	0.568	0.639	0.755	0.743	0.599	0.557	0.536	0.597	0.638	0.635	0.560	0.553		(40)
(41)	Clear-Sky Solar Irradiance	taud	1.807	1.700	1.520	1.585	1.918	2.038	2.094	1.909	1.792	1.742	1.895	1.863		(41)
		Ebn at Noon	683	666	614	636	733	758	774	729	688	659	686	676		(42)
		Edh at Noon	195	231	289	275	196	172	163	196	217	218	177	179		(43)
(44)	All-Sky Solar Radiation	RadAvg	3.09	3.22	3.53	3.81	4.36	4.57	5.51	5.09	4.80	4.33	3.45	3.13		(44)
		RadStd	0.44	0.55	0.49	0.61	0.62	0.61	0.45	0.47	0.34	0.49	0.38	0.42		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.34	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	+138	+98	(46)	
(47)	Regional (4 neighbors)	+0.27	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	+85	+79	(47)	

Nomenclature: See separate page